

# Application of Physics-Informed Neural Networks with Topological Validation on Gene Regulatory Networks

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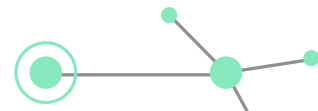
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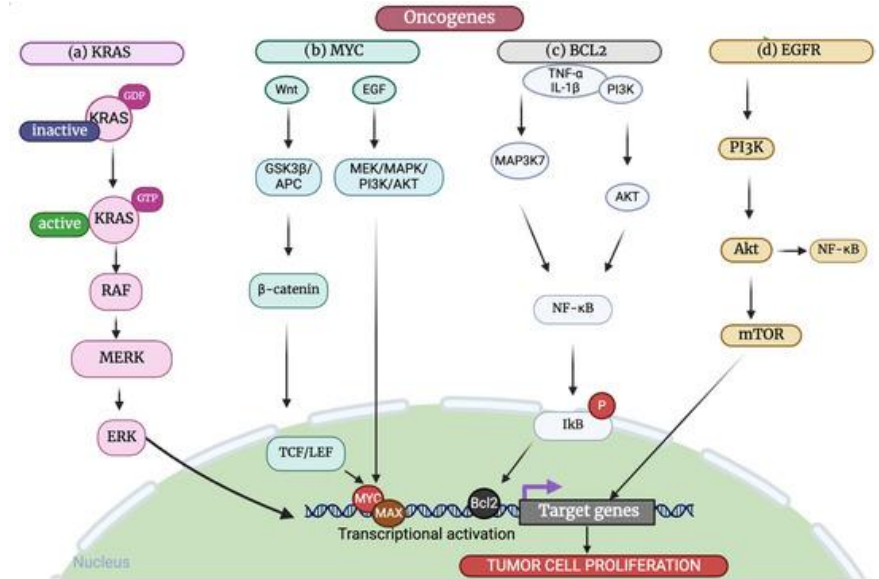
**04** Results and insights to Epidemiology and Ecology



# Cause of Cancer:

- 1) Tobacco
- 2) Infections
- 3) Radiation
- 4) Exposure to chemicals
- 5) Diets

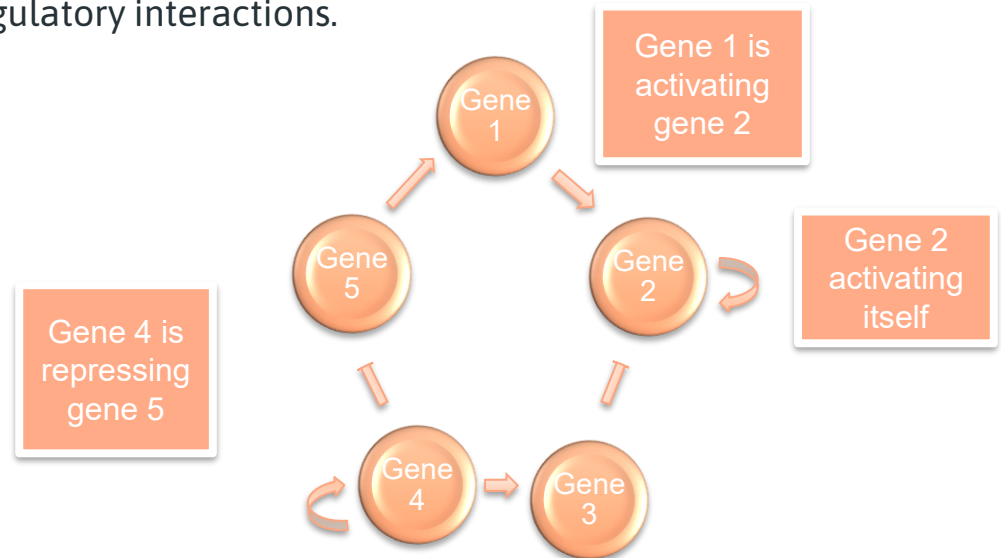
**6) Genes!**



# Gene Regulatory Networks:

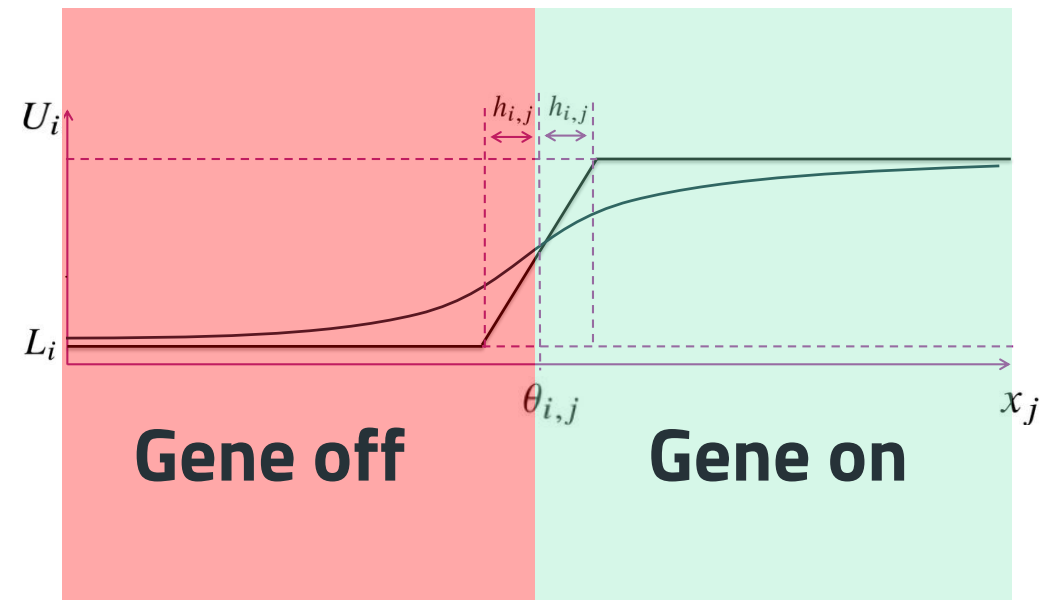
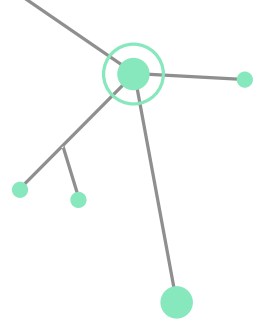
A **gene regulatory network (GRN)** is a representation of how genes control each other's activity through regulatory interactions.

Made of nodes and edges.



# Gene-Gene Interactions

The effect of one gene on another is observed only when the relevant expression level crosses a threshold  $\theta_{i,j}$ . These interactions can be modeled using sigmoidal functions including:



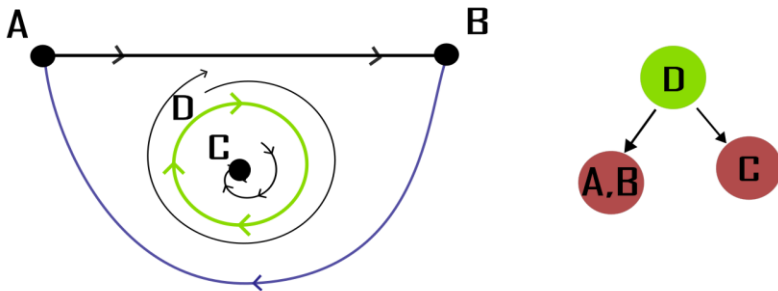
$$g_i(x_j) = \begin{cases} U_i, & x_j > \theta_{i,j} + h_{i,j} \\ S_{i,j}(x_j), & \theta_{i,j} - h_{i,j} \leq x_j \leq \theta_{i,j} + h_{i,j} \\ L_i, & x < \theta_{i,j} - h_{i,j}, \end{cases}$$

$$\frac{dx_j}{dt} = \text{Production}_j(x_i(t)) - \text{Decay}_j(x_j(t))$$

# Morse Decomposition:

Let  $X$  be a compact metric space and  $\varphi: \mathbb{R}_{\geq 0} \times X \rightarrow X$  a continuous semiflow.

A family  $\mathcal{M} = \{M_\alpha\}_{\alpha \in A}$  of nonempty, pairwise disjoint, compact, invariant sets is a **Morse decomposition** of  $X$  if there is a strict partial order  $<$  on  $A$  such that for every  $x \in X$  whose full orbit is defined forward and backward, either  $x \in \bigcup_\alpha M_\alpha$  or there exist indices  $\alpha \leq \beta$  with  $\omega(x) \subset M_\alpha$ ,  $\alpha \leq \beta$ ,  $M_\beta \subset \alpha(x)$ , where  $\omega(x)$  and  $\alpha(x)$  are the omega and alpha limit sets of  $x$ . All recurrent dynamics is contained in the  $M_\alpha$

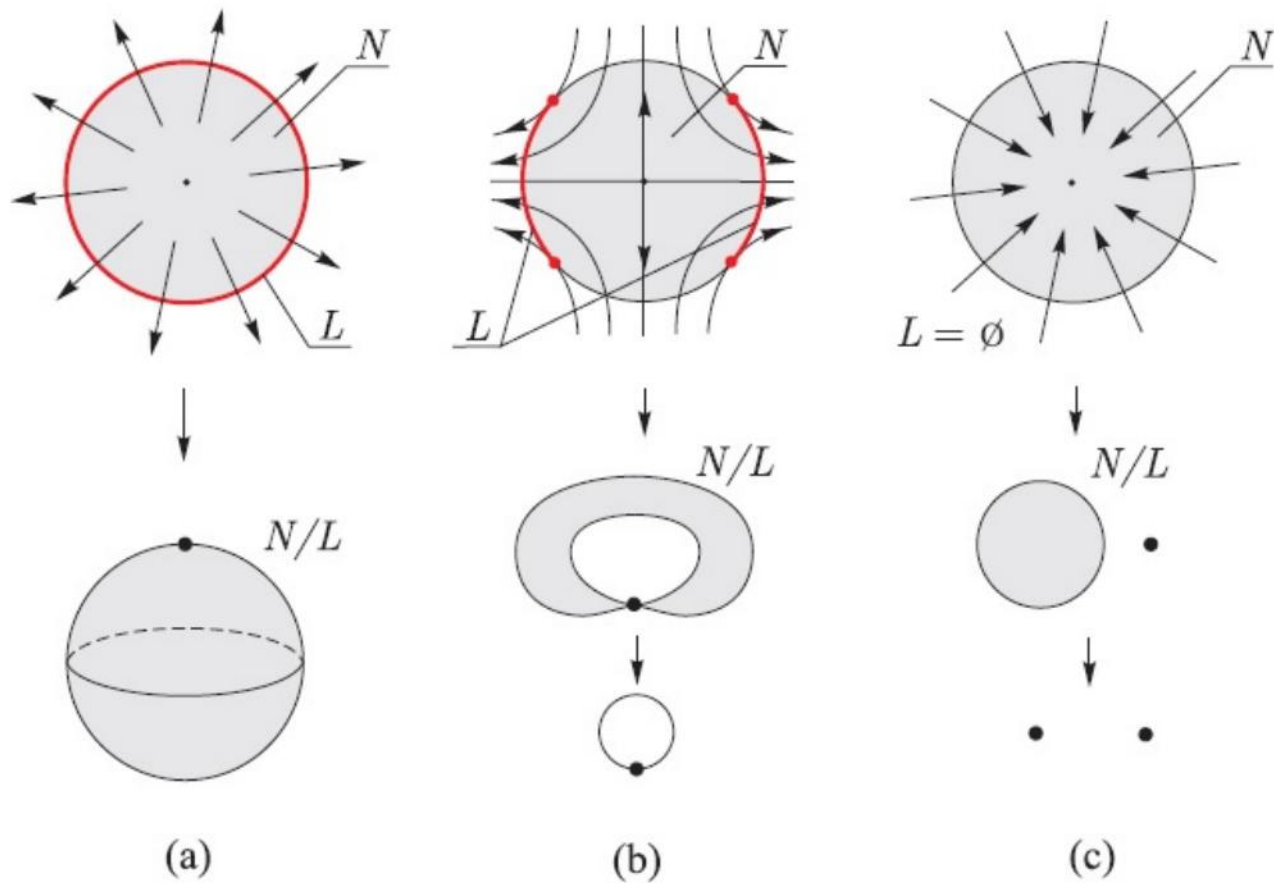


Let  $S$  be an isolated invariant set. A pair of compact sets  $(N, L)$  with  $L \subset N$  is an index pair for  $S$  if:  
Index pair conditions

1. Isolation of  $S$ :  $S = \text{Inv}(\text{cl}(N \setminus L))$ ,  $N \setminus L$  is a neighborhood of  $S$ .
2. Positive invariance of  $L \subset N$ : if  $x \in L$  and  $\varphi([0, t], x) \subset N$ , then  $\varphi([0, t], x) \subset L$ .
3. Exit set property: if  $x \in N$  and  $t_0 > 0$  with  $\varphi(t_0, x) \notin N$ , then there exists  $0 \leq t_1 < t_0$  such that  $\varphi(t_1, x) \in L$ .

The homotopy Conley index of  $S$  is the homotopy type of the pointed space obtained by collapsing  $L$  to a point:

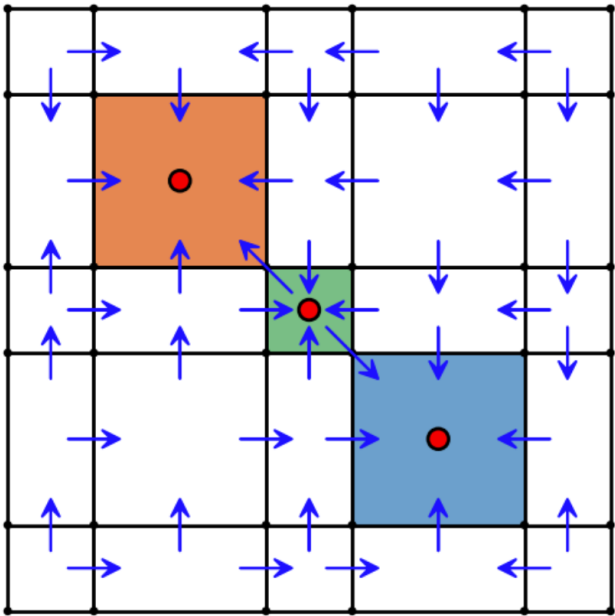
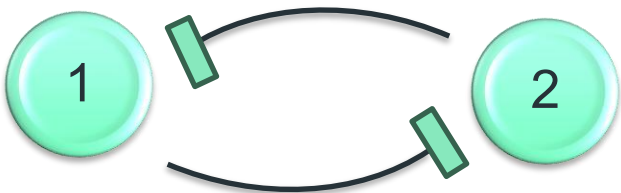
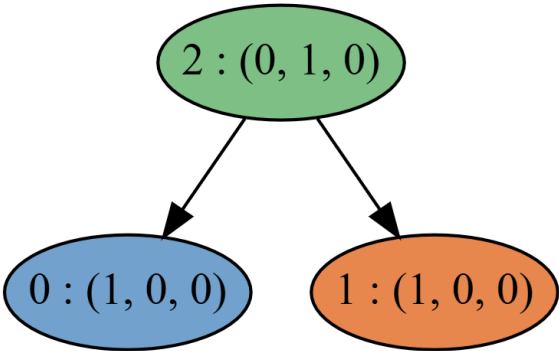
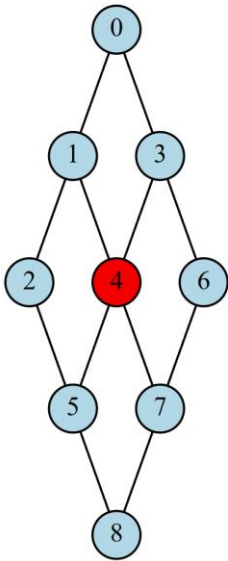
$$h(S) \sim (N/L, [L]).$$



The Conley index for a source, a saddle and a sink.

<https://mat.uab.cat/~sytm/pdf/sytm-Barge.pdf>

# Example 1: Toggle switch



# Problem Statement

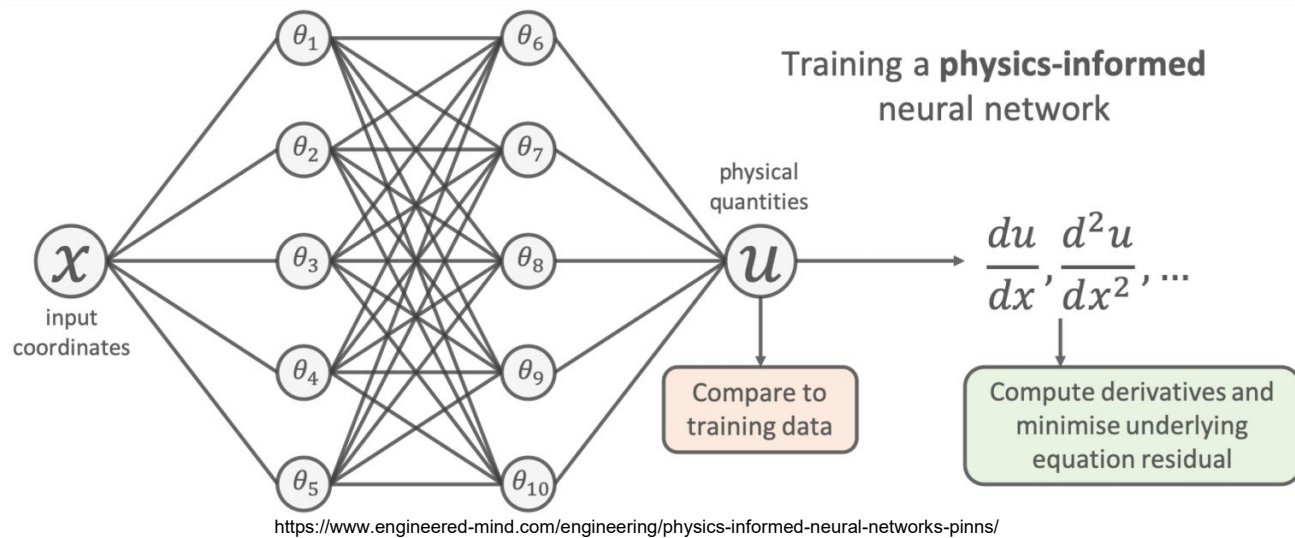
Biological data

Learn  
parameters  $L, U, \theta, d$

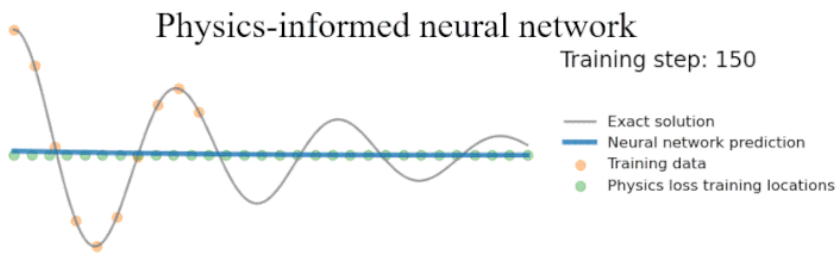
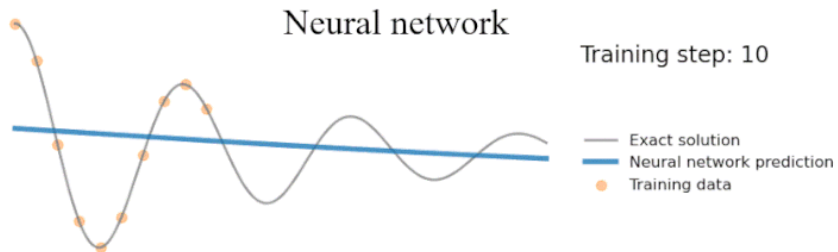
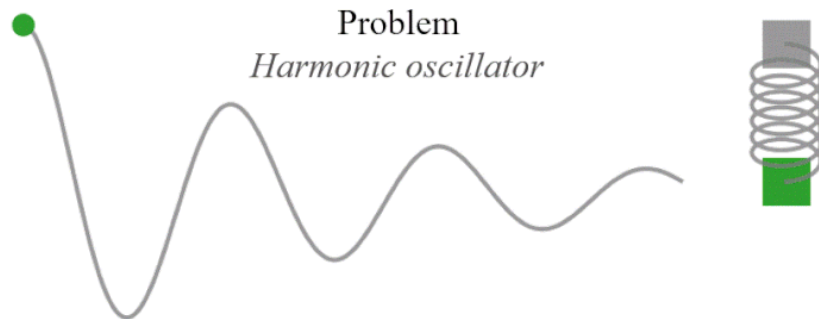
Get Dynamics  
through DSGRN



# PINN: Physics informed Neural Networks



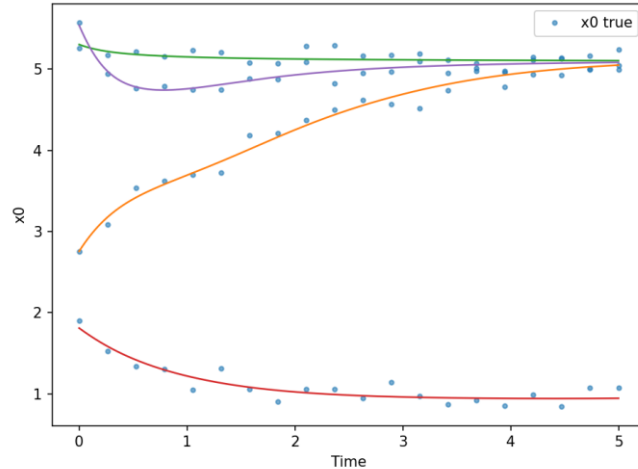
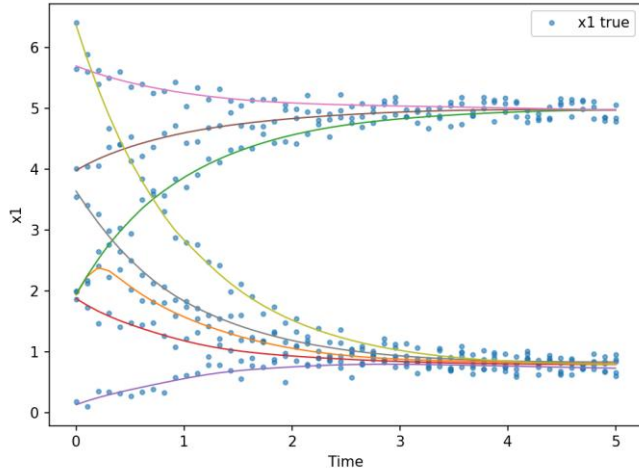
$$\mathcal{L}(w, b) = \underbrace{\frac{1}{N_d} \sum_{i=1}^{N_d} (F_{w,b}(x_i) - y_i)^2}_{\text{data term}} + \lambda_f \underbrace{\frac{1}{N_c} \sum_{j=1}^{N_c} \left( \mathcal{N}[F_{w,b}](x_j^{(c)}) \right)^2}_{\text{physics residual}} + \lambda_b \underbrace{\frac{1}{N_b} \sum_{k=1}^{N_b} \left( \mathcal{B}[F_{w,b}](x_k^{(b)}) \right)^2}_{\text{BC/IC term}}.$$



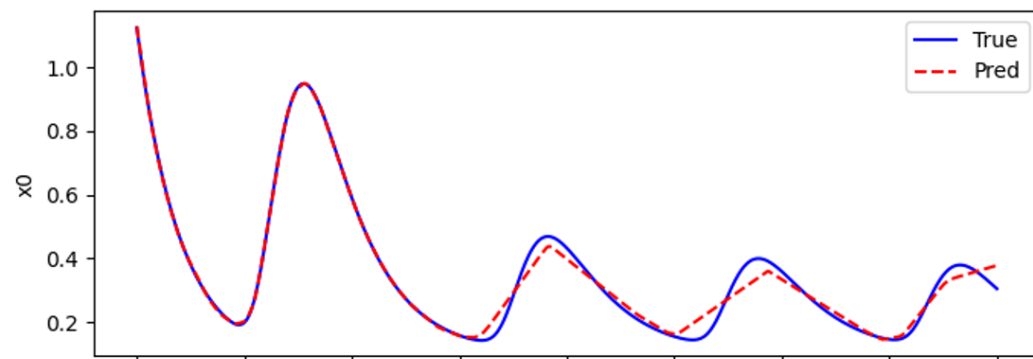
# Inverse Problem:

Can we estimate parameters inside our differential equations? What is the benefit of using PINN compared to least square?

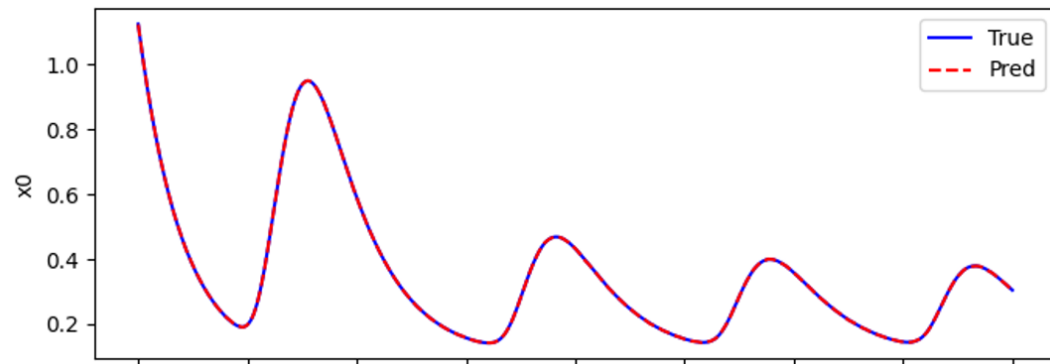
YES

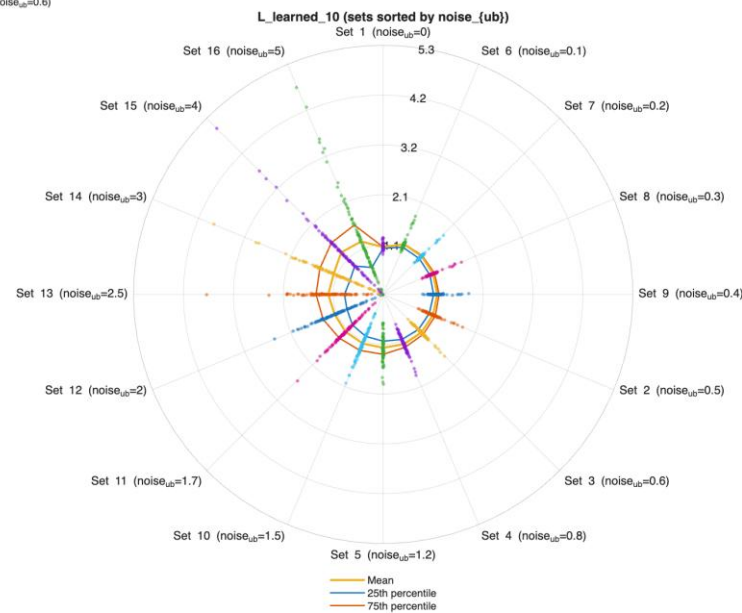
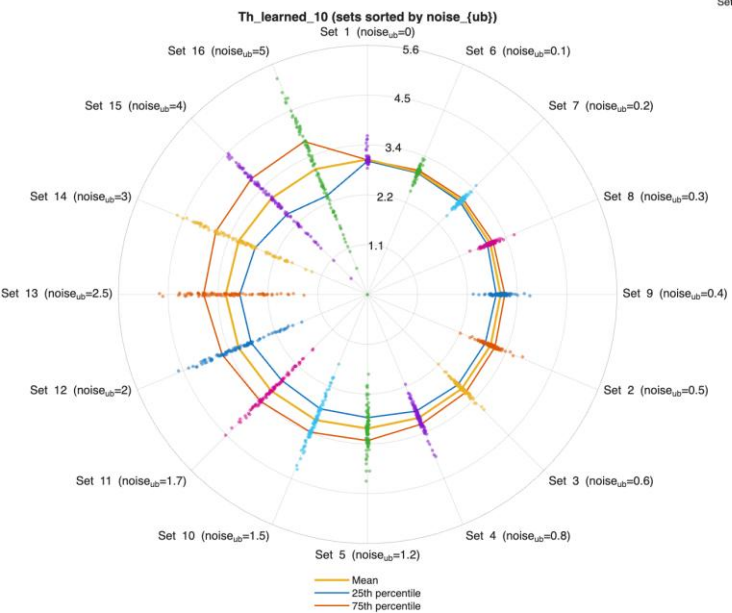
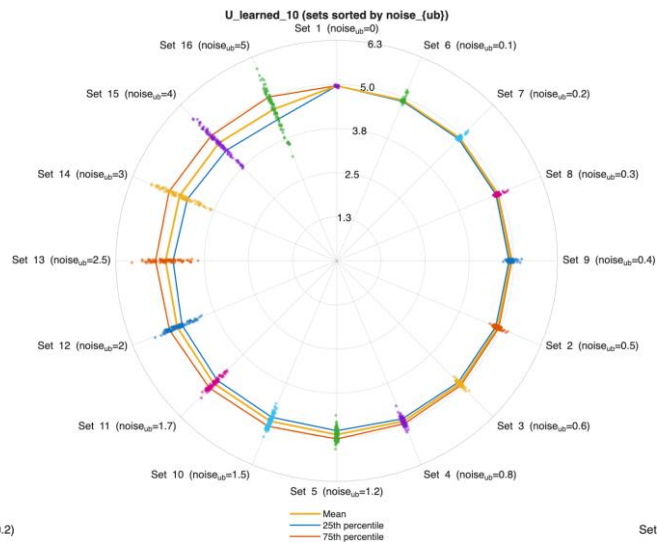


No feature map used

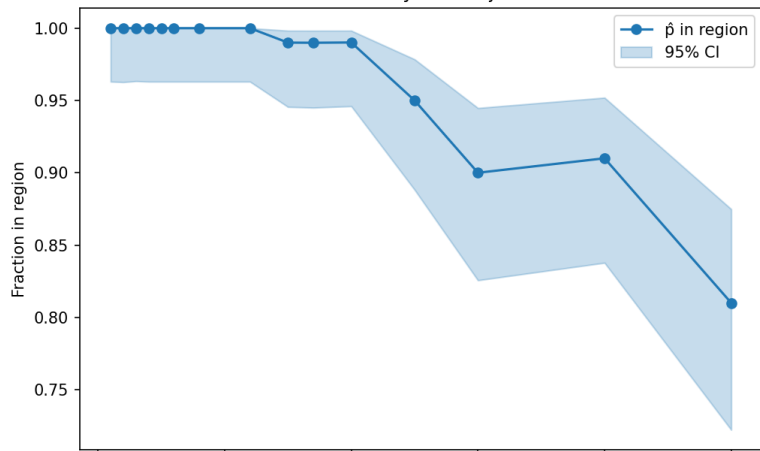


Feature map applied

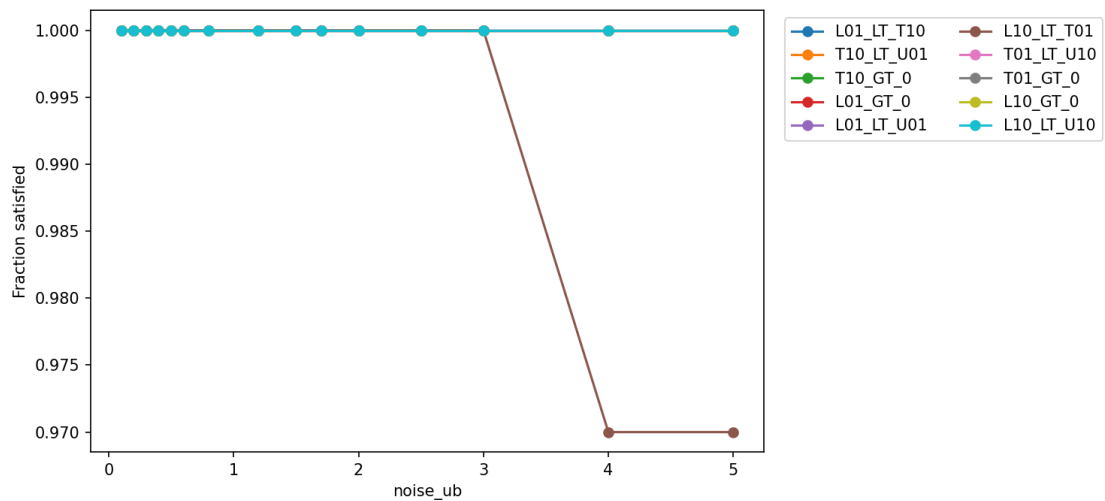
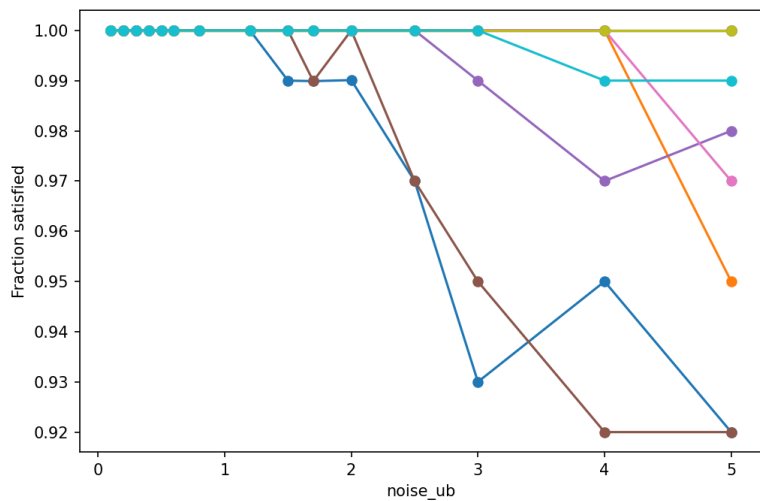
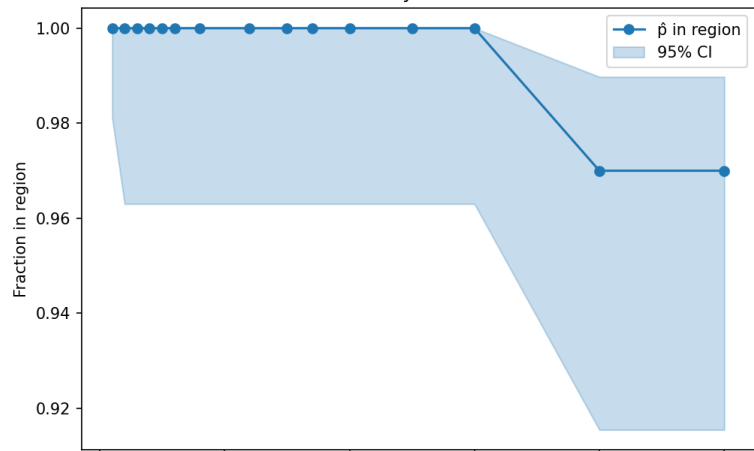




Summary: diff Traj IC



Summary: diff NN IC





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# Thanks!

**Do you have any questions?**

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# References

- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019).** *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations.* **Journal of Computational Physics**, **378**, 686–707.
- Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021).** *Physics-informed machine learning.* **Nature Reviews Physics**, **3**, 422–440.
- Conley, C. (1978).** *Isolated Invariant Sets and the Morse Index.* **CBMS Regional Conference Series in Mathematics, Vol. 38.** American Mathematical Society.
- Mischaikow, K., & Mrozek, M. (2002).** *Conley index theory.* In: **Handbook of Dynamical Systems, Vol. 2**, 393–460. Elsevier.
- Glass, L., & Kauffman, S. A. (1973).** *The logical analysis of continuous, non-linear biochemical control networks.* **Journal of Theoretical Biology**, **39**, 103–129.
- Cummins, B., Gedeon, T., Harker, S., Krause, R., Mischaikow, K., Mok, K., & Yoshida, A. (2016).** *DSGRN: Dynamic Signatures Generated by Regulatory Networks.* (Foundational DSGRN framework paper / project reference).